Gurukiran Reddy Kasireddy Software Engineer

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Summary

Backend Software Engineer with 3 years of experience building high-scale backend services, distributed systems, and cloud-native applications in healthcare and enterprise environments. Strong expertise in Golang, Python (Django, Flask), Node.js, and AWS, with hands-on experience designing microservices, distributed cache systems, message queue-based architectures, and scalable REST/GraphQL APIs. Experienced in integrating AI-driven analytics and machine learning components into backend systems to enable intelligent data processing and automation. Proven track record of optimizing large-scale data pipelines, enhancing system performance, and delivering production-ready systems with strong system design fundamentals.

Skills

- **Programming Languages:** Golang (Go), Python, Java, JavaScript, SQL, Unix Shell Scripting
- **Backend & Architecture:** Django, Flask, Node.js, Express.js, REST APIs, GraphQL, Microservices Architecture, Distributed Systems, Message Queue-Based Architecture, Event-Driven Systems, Scalable HTTP Services
- **Frontend Technologies:** AngularJS, React, Vue.js, Bootstrap
- **Data Engineering & Processing:** PySpark, Pandas, NumPy, SQLAlchemy, ETL Development, Data Validation, Large-Scale Data Processing
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Cassandra, Distributed Cache Systems, Database Optimization & Indexing
- **Cloud & DevOps:** AWS (EC2, ECS, EKS, Lambda, RDS, S3), Docker, CI/CD (Jenkins, AWS CodeBuild, CodeDeploy), Git
- **Tools & Practices:** Agile (Scrum), JIRA, Azure DevOps, Structured Logging & Monitoring

Experience

UnitedHealth Group, Software Developer — 08/2025 - Present | USA

- Developed high-scale backend services in Golang and Python supporting 12K+ daily healthcare transactions, reducing manual processing effort by approximately 40% and improving system reliability.
- Architected and implemented AWS-based microservices (ECS, Lambda, RDS) processing 8M+ structured and unstructured records per batch cycle, improving throughput and maintaining fault tolerance.
- Optimized large-scale PySpark pipelines handling multi-million record healthcare datasets, reducing end-to-end processing time by 30% through parallel execution tuning and query refinement.
- Improved API response latency from ~450ms to ~250ms by refining SQL indexing strategies and optimizing query execution plans in PostgreSQL and MySQL.
- Integrated AI-driven analytics components into backend workflows, enabling automated decision support and reducing manual review cycles by approximately 25%.
- Applied distributed cache system concepts to minimize redundant database calls and improve API response times for frequently accessed datasets.
- Developed secure, scalable APIs supporting interactive dashboards built with React and AngularJS, enabling near real-time reporting and operational visibility.
- Deployed and managed containerized services using Docker and AWS infrastructure, ensuring high availability and streamlined CI/CD cycles.
- Leveraged ML/AI libraries (Pandas, NumPy, Scikit-learn) to support internal analytics and data-driven insights while maintaining production-grade backend engineering standards.

BGCCCI Bronx Community College CUNY, Software Intern – Research Associate — 09/2024 - 12/2024 | USA

- Engineered Python-based ETL pipelines processing 3M+ records, improving data validation accuracy and reducing inconsistencies by 15%.
- Built backend processing services integrated with analytics dashboards, enhancing workflow transparency and reporting accuracy across research operations.
- Containerized backend preprocessing workflows using Docker, decreasing environment setup time by 50% and improving reproducibility.
- Contributed to a system design heavy project focused on modular backend architecture for research data automation and structured reporting pipelines.

CitiusTech Inc, Software Developer — 05/2021 - 07/2023 | India

- Designed and developed high-scale backend services using Django, Flask, and Node.js, improving application performance by 25% through optimized database schema design and efficient ORM usage.
- Implemented message queue-based microservices on AWS (Lambda, ECS) to handle 10K+ daily events, reducing synchronous processing load and improving scalability.
- Built secure and high-performance RESTful and GraphQL APIs with JWT authentication, input validation, and structured error handling to support dynamic frontend and third-party integrations.
- Developed server-side backend components enabling seamless integration with React and Vue-based frontend applications.
- Designed distributed backend components integrating Cassandra and PostgreSQL for high-throughput data storage, retrieval, and analytics support.
- Automated Cassandra database operations and ETL workflows using Python and Node.js, improving data validation, monitoring, and deployment integration.
- Built and optimized enterprise data processing pipelines using Pandas, NumPy, and Scikit-learn to support backend analytics and intelligent automation features.
- Orchestrated containerized deployments using Docker and AWS ECS/EKS, ensuring infrastructure consistency and scalable cloud-native deployments.
- Implemented and maintained CI/CD pipelines using Jenkins, AWS CodeBuild, CodeDeploy, and Git workflows to enable automated testing and reliable production releases.
- Participated in Agile development cycles, contributing to sprint planning, code reviews, production debugging, and system design discussions for scalable enterprise applications.

Education

New Jersey Institute of Technology, Master of Science in Computer & Information Systems — 09/2023 – 05/2025 | Newark, NJ

PES University, Bachelor of Engineering in Computer Science and Engineering — 08/2018 – 05/2022 | Bangalore, India